Akoizo Original Practice - Form 4 - Answer key

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A proportional controller sets motor power based on:

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

A. The elapsed time only

B. The pusher width

C. The remaining distance to the target

D. The battery's label

Correct: C

2. (2 points) Target Distance 6.00 m and Bottle Line Distance 1.50 m. Where is the Bottle Line from the Start Point?

A. 1.50 m

B. 6.00 m

C. 7.50 m

D. 4.50 m

Correct: C

3. (2 points) Which action results in a competition violation?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

A. Following the vehicle down the track before being called

B. Measuring the track during event time

C. Cleaning the track while keeping it dry

D. Programming over USB

Correct: A

4. (2 points) A pusher's widest opening is 35.0 cm. What is its Pusher Bonus?

A. 0

B. -52.5

C. +35

D. -35

Correct: A

5. (4 points) Why calibrate distance per count with a measured run?

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

answer: Actual wheel diameter, tire compression, and slip differ from nominal values, so a measured run gives the true conversion. (4 points).

6. (4 points) A vehicle pauses 2 s after the pencil trigger before moving. Does the pause count?

answer: Yes. Run Time starts at the trigger, and any pause before moving is included. (4 points).

7. (4 points) Describe a good Measurement Point.

Unit: Power, construction limits, and legal parts.
Worked-case data: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

answer: A rigid pointed tip, such as a skewer end, that is the rearmost part, at most 1.0 cm above the track, and easy to reach. (4 points).

8. (4 points) Why test at extreme combinations of posted values?

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

answer: Long distances with short times demand top speed, while short paths with long times demand slow, steady control; a program must work at both extremes. (4 points).

9. (4 points) How does sideways drift affect the score even if the forward distance is perfect?

answer: Vehicle Distance is measured point to point, so lateral offset adds to the distance; drift can also cause the bottle to be missed or pushed off line. (4 points).

10. (4 points) Why is testing on the competition floor beforehand restricted?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

answer: Rolling on the track gives an unfair calibration advantage, so it requires tournament permission and is banned during competition outside runs. (4 points).

11. (6 points) Solve this case and explain your reasoning: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).

reasoning: 9.0 cm bonus: (9 - 35) x 1.5 = -39.0; 15.0 cm bonus: -30.0.
Expected miss cost: 9.0 cm -> 0.2 x 420 = 84; 15.0 cm -> 0.
Expected net: 9.0 cm about +45; 15.0 cm about -30. (3 points).

conclusion: The reliable 15.0 cm pusher is about 75 points better on average, despite its smaller bonus. (3 points).

12. (2 points) What should the team do?

Supplied synthetic observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

A. Switch to time-based stopping

B. Use separate calibration factors for forward and reverse

C. Ignore the error because it is under 1 percent

Correct: B

13. (2 points) Define “Measurement Point (MP)” in the context of Electric Vehicle.

answer: Rearmost point of the vehicle, at most 1.0 cm above the track, used for distance. (2 points).

14. (2 points) Define “AA battery” in the context of Electric Vehicle.

answer: Commercial labeled cell, 1.2-1.5 V when charged; up to eight allowed. (2 points).

15. (2 points) Define “Closed-loop control” in the context of Electric Vehicle.

answer: Commands corrected using sensor feedback. (2 points).

16. (2 points) Define “Bottle Location” in the context of Electric Vehicle.

answer: Bottle start on the center line: 2, 3, or 4 m by tournament level. (2 points).

17. (2 points) Define “Ready-to-run” in the context of Electric Vehicle.

answer: Vehicle placed at the start, waiting for the trigger. (2 points).

18. (2 points) Define “Pocket” in the context of Electric Vehicle.

answer: The shaped area where the captured bottle rests. (2 points).